
Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

1 A printed claim that the scorecaster breaks online conformal coverage meets its
2 refutation by overlapping authors; this paper joins them and locates the edge
3 of the set that refutation relies on. Conformal PID states long-run coverage for
4 any bounded scorecaster; a corollary keeps it for predictable perturbations of the
5 deployed threshold inside an admissible radius. The claim was acted on; the
6 readout’s placement and the corollary’s derivation are credited by name. At the null
7 scorecaster and the minimal admissible saturator that radius is *tight*. A legal dead
8 band on the *completed* threshold exists that set at its radius; past it miscoverage is
9 1.000000 against $\alpha = 0.1$: coverage fails outright, not merely its rate. The edge
10 belongs to the saturator–scorecaster pair, not the readout; elsewhere the condition
11 is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band covers, as
12 measured; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

13 1 Introduction

14 A refereed paper states that a scorecaster breaks Conformal PID’s coverage guarantee. A later one,
15 by the same first and last authors, proves in the same notation that a bounded perturbation of the
16 deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the
17 joining is what this paper adds.

18 An online conformal method [Angelopoulos and Bates, 2023] moves a threshold each round. Forecast
19 stability names, scores and prices that movement [Tunc et al., 2013, Godahewa et al., 2025, Van Belle
20 et al., 2023, Pritularga and Kourntzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying
21 temporal structure at fixed level came earlier, in forecast verification, matching predictive marginals
22 on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. Coverage
23 and mean length under-describe a deployed interval [Min et al., 2026, Vaze, 2026]. None of it is
24 claimed here.

25 **The record carries a claim and its refutation; this paper joins them.** Conformal PID [An-
26 gelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be
27 predictable and lie in $[-b/2, b/2]$: “any function of the past” (Theorem 1). Such a readout is already
28 quantified over. The claim reads “[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical
29 coverage guarantee” [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, ver-
30 batim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2]
31 add any predictable d_t , $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ with $\mu_t \in [0, 1]$ predictable, to the deployed threshold;
32 since $\hat{q}_t + d_t$ still lies in $[-b/2, b/2]$ it is itself a legal scorecaster, so the result “has exactly the CPID
33 error-integration form” and $|E_T| = o(T)$. The same generic-slot argument carries AcMCP’s long-run
34 coverage [Wang and Hyndman, 2024]. Both identifiers and “scorecaster” were searched across arXiv
35 full text, arXiv abstracts, OpenReview, and Semantic Scholar’s six citers of Li and Rodríguez [2025]:
36 none returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice “arbitrary”
37 and lacking “principled guidance” while asserting valid coverage: model selection, not validity. **We**
38 **claim neither the placement nor the derivation.**

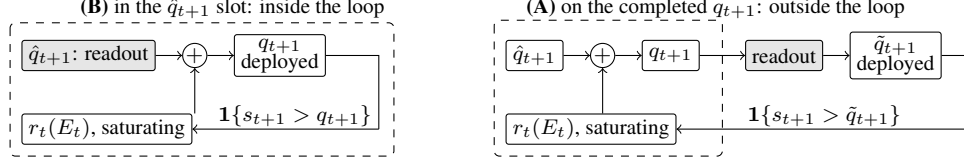


Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In (B) the loop closes on the integrator’s own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster’s range, inside Corollary 2’s admissible set. In (A) it closes on $\tilde{q}_{t+1}$, which it never computed, and nothing keeps the deployed sequence inside that set.

39 **Where Corollary 2 stops, its radius condition is tight.** It is a *retention* result, silent outside that
40 radius, and the question it leaves is whether the radius is sharp. Section 3 exhibits a legal perturbation
41 leaving it: the L_1 dead-band family characterised here on the *completed* threshold, whose edge
42 coincides with the corollary’s radius at the null scorecaster. Past it coverage fails outright, not merely
43 its rate: the published sufficient condition is necessary there. Appendix C sets both claims beside
44 four vocabularies for the same movement.

45 2 Setup

46 **The producer.** A forecaster fit before t induces a score s_t , and a threshold q_t is emitted before
47 y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\text{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The
48 target is long-run coverage, $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID
49 [Angelopoulos et al., 2023] manipulates the threshold on the score scale:

$$q_{t+1} = \hat{q}_{t+1} + r_t \left(\sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

50 Angelopoulos et al.’s iteration (5). Its r_t saturates: condition (4) requires $|r_t(x)| \geq b$ with the sign of
51 x once $|x| \geq c h(t)$, for h nonnegative, nondecreasing and sublinear. With $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$,
52 Proposition 2 gives $|E_T| \leq c h(T) + 1$, read on realised scores.

53 **One scalar, two readouts, two places.** Damping a sequentially updated scalar has two standard
54 readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1 - w)\hat{q}_{t+1}^{\text{raw}} + w\tilde{q}_t$
55 [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied
56 to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band,
57 moving only when the target is more than τ away, then pulling back exactly τ [Constantinides,
58 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; (A),
59 Placement A, is measured here. *The dead band (L_1) is not a stylistic alternative to the partial-*
60 *adjustment (L_2) form; it is the family whose failure Section 3 characterises. Neither is safe by*
61 *construction: L_2 forfeits Proposition 2’s rate while covering, L_1 past the radius loses coverage, and*
62 *at the legal $\hat{q} \equiv -b/2$ both lose it.*

63 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*, $\mathcal{T}_H =$
64 $\sum_{t=2}^H |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new, and the name is
65 a convenience rather than a contribution, used here for one reading only: Appendix C names ten for
66 the same movement across four vocabularies.

67 **Harness.** The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b \text{clip}(x/(c h(t)), -1, 1)$,
68 with $c = 1$ and $h(t) = \log(t + 2)$, meets (4) with equality. The scorecaster is null and legal, reducing
69 (1) to the pure error integration Proposition 2 addresses, so the readout is the only manipulated
70 variable. Neither choice is neutral: (4) is a lower bound met exactly here, and Theorem 1 permits
71 scorecasters $b/2$ either side of null, predictable but not required to be constant in t . The adversary is
72 specified, not tie-broken. It plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the
73 $\varepsilon \downarrow 0$ reading of “sets the score at the threshold”, firing the strict indicator, legal on both sides of
74 Theorem 1’s $[-b/2, b/2]$, so $b/2 + b/2 = b$ is exactly tight. *That specification names one member*
75 *of a class: any adapted $s_t \in [-b/2, b/2]$ is legal, the constant $s_t \equiv -b/2$ included. Every number*
76 *in Table 1 (App. A) is measured against the one adversary above; Section 3’s two boundaries are*
77 *quantified over the class, and they differ.*

78 3 The edge of the admissible set, located and proved

79 **The retention result is published, and tested from outside.** The radius restated in Section 1 is
 80 the one Li et al. [2025, Corollary 2] prove, and it *retains* $|E_T| \leq ch(T) + 1$: the radius is theirs. A
 81 downstream partial adjustment of weight w leaves it; Table 1 (App. A) prices that departure, and five
 82 of its eleven arms exceed the bound at $T = 10^6$. The forfeit is a property of the smoothing *strength*,
 83 not Placement A, and it buys something: travel falls from 28,311 to 202. A readout on the completed
 84 q_{t+1} touches neither r_t nor the integrator’s argument, so it cannot put (4) at risk. It delays a correction
 85 already computed, so smoothing the output does not damp E_t but feeds it.

86 **The L_1 dead band leaves the radius, and the edge is a theorem.** The dead band of Section 2 keeps
 87 $|\hat{q}_{t+1} - q_{t+1}| \leq \tau$ at every round, so “inside Corollary 2” reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation
 88 the deployed value is pinned τ below the integrator’s ceiling. Outside the radius two levels decide the
 89 answer and condition (4) names only one of them. Write $A_t^+ = \sup_x r_t(x)$ and $A_t^- = \inf_x r_t(x)$ for
 90 the integrator’s extremes, and $\Lambda_t^+ = \inf_{x \geq ch(t)} r_t(x)$, $\Lambda_t^- = \sup_{x \leq -ch(t)} r_t(x)$ for the levels it is
 91 *guaranteed* to reach once the accumulator clears the radius. Condition (4) says exactly $\Lambda_t^+ \geq b$ and
 92 $\Lambda_t^- \leq -b$; that $\Lambda_t^\pm = A_t^\pm$, that the integrator reaches its extremes where (4) puts them, is a further
 93 hypothesis and not one (4) supplies. Set

$$\tau^{*\pm} = \sup_t (|A_t^\pm| \pm \hat{q}_{t+1}) - b/2, \quad \sigma^\pm = \inf_t (|\Lambda_t^\pm| \pm \hat{q}_{t+1}) - b/2. \quad (2)$$

94 **Failure, over the whole admissible class.** If $\tau > \tau^{*+}$ then $\hat{q}_{t+1} + A_t^+ - \tau < b/2$ at every round,
 95 so $\tilde{q}_{t+1} \leq \max(\tilde{q}_t, q_{t+1} - \tau)$ makes $\{\tilde{q} < b/2\}$ absorbing: a release upward lands below $b/2$, and a
 96 hold or a release downward cannot lift the deployed value at all. Against $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ the
 97 indicator is then $\mathbf{1}\{\tilde{q}_t < b/2\}$ and every round misses. If $\tau \geq \tau^{*-}$ the constant legal play $s_t \equiv -b/2$
 98 makes $\{\tilde{q} \geq -b/2\}$ absorbing by the mirror inequality, so no round can miss and the realised rate
 99 is 0 rather than α . Either way long-run coverage is lost, with $|E_T|$ linear in T . Nothing here asks $\hat{q}$
 100 to be constant, asks the extremes to be attained anywhere, or asks $A_t^\pm$ to be finite beyond what the
 101 hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If $\tau \leq \sigma^+$
 102 then $E_{t-1} \geq ch(t-1)$ forces $q_t \geq b/2 + \tau$ and so $\tilde{q}_t \geq b/2$, whence $\text{err}_t = 0$ against *every* legal
 103 adversary; if $\tau < \sigma^-$ the mirror forces $\text{err}_t = 1$ below $-ch(t-1)$. The accumulator cannot then
 104 leave $[-ch(t), ch(t)]$ by more than one round, and $|E_T| \leq ch(T) + 1$: Proposition 2’s own bound,
 105 retained verbatim rather than weakened to $o(T)$. Using condition (4) alone this already certifies
 106 $\tau \leq b/2 + \inf_t \hat{q}_t$ and $\tau < b/2 - \sup_t \hat{q}_t$, which for a constant scorecaster is Corollary 2’s radius at
 107 $\mu_t = 1$.

108 **The two directions meet, with no gap, on the sub-class the five settings inhabit.** If $\hat{q}_t \equiv \hat{q}$, if
 109 $\Lambda_t^\pm = A_t^\pm$, and if neither depends on t , then $\sigma^\pm = \tau^{*\pm}$ and **coverage is retained against every legal**
 110 **adapted adversary if and only if $\tau \leq \tau^{*+}$ and $\tau < \tau^{*-}$.** For a symmetric saturator that is one line:
 111 coverage if and only if $\tau \leq \sup_x r_t(x) - b/2 - |\hat{q}|$, with the endpoint included only when $\hat{q} < 0$.
 112 Every setting run here is in that sub-class, so the sharp form applies to all five. **What was printed**
 113 **before is the failure half.** τ^{*+} is the rule Figures 2 and 3 (App. A) draw, and all 19 dead-band runs
 114 agree with it read against the adversary the harness plays. It is not a retention condition, it omits τ^{*-}
 115 entirely, and Table 2 (App. A) separates the two with legal objects only. The supremum can sit where
 116 the dynamics never go: since $|E_t| \leq (1 - \alpha)t$, a saturator equal to the harness’s inside $|x| \leq t^2$ and to
 117 $\pm 10b$ outside it satisfies (4), reads $\tau^{*+} = 19$ and fails at $\tau = 1.001$; and a legal time-varying $\hat{q}$ splits
 118 $\sup_t$ from $\inf_t$, $\hat{q}_t = b/2$ on t a power of two and 0 elsewhere reading $\tau^{*+} = 2$ against $\sigma^+ = 1$.

119 **The sharp form is one number, and its hypotheses are what buy it.** Specialise to a symmetric
 120 saturator that meets (4) with equality and attains its extremes there ($\Lambda_t^\pm = A_t^\pm = \pm b$; the harness’s
 121 $r_t = b \text{ clip}(x/(ch(t)), \pm 1)$ is one, Section 2), and to a constant $\hat{q}$. Then

$$\min(\tau^{*+}, \tau^{*-}) = b/2 - |\hat{q}|, \quad (3)$$

122 with the endpoint admissible only when $\hat{q} < 0$: exactly Corollary 2’s radius at $\mu_t = 1$, now as an
 123 if-and-only-if. **The necessity direction above assumes none of it:** $\tau > \tau^{*+}$, or $\tau \geq \tau^{*-}$, forfeits
 124 coverage against every admissible r_t and every legal predictable $\{\hat{q}_t\}$, extremes attained or not, and
 125 that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather
 126 than decorative: the unattained supremum and the varying- $\hat{q}$ split two paragraphs above are exactly

127 what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they
128 always do.

129 **The mirror boundary is already measured, on this paper’s own grid.** The τ -cliff run’s exact-
130 threshold column, read against the mirror adversary $s_t \equiv -b/2$ (Figure 2, App. A), locates $\tau^{*-} = 1$
131 at the null scorecaster, closed on the failing side, exactly where (2) puts it. Both 1.000000 and
132 0.000000 are failures of the same order, $|E_T| = (1 - \alpha)T$ and αT , **so the failure criterion is $|E_T|$,**
133 **not the miscoverage rate;** and once $\hat{q}$ varies the rate stops being a tell at all.

134 **Both degenerate edges, and what they need.** An unbounded supremum is not on its own enough
135 for $\tau^{*+} = \infty$ to mean what it looks like: the same unreachable-set construction with $r_t(x) = x$
136 past $|x| > t^2$ has no finite supremum and fails at $\tau = 1.5$. What the tangent family has, and that
137 construction does not, is a genuine reach: $\Lambda_t(X) = \tan(\arctan(b)X/(ch(t)))$, so the accumulator
138 level at which the deployed threshold is forced to $b/2$ is $\kappa(\tau)ch(t)$ with $\kappa(\tau) = \arctan(b/2 +$
139 $\tau - \hat{q})/\arctan(b)$, finite at every finite τ . Coverage is retained at every width, as measured; but
140 $\kappa(\tau) > 1$ once $\tau > b/2 + \hat{q}$, so Proposition 2’s own *constant* is forfeited past $\tau = b/2$ while coverage
141 is not. At $\tau = 1.5$, $T = 10^6$ the excursion is 15.10 against Proposition 2’s 14.8155 and against
142 $\kappa(1.5)ch(T) + 1 = 15.8530$. At the other edge, $\sup_t \hat{q}_t = -b/2$ forces $\hat{q} \equiv -b/2$, since every $\hat{q}_t$
143 is at least $-b/2$ already. There $q_{t+1} - \tau \leq -b/2 + b - \tau < b/2$ for every $\tau > 0$ and every t , and
144 $\hat{q}_1 = 0 < b/2$, so the failing set is absorbing from round 1 with no transient: the admissible window
145 is the single point $\tau = 0$, and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$.

146 **What that establishes is that the published condition is tight.** At $\mu_t = 1$, $\hat{q} \equiv 0$, $\tau = 0.9$ covers at
147 0.100012 while $\tau = 1.5$ returns 1.000000 with $\max_t |E_t| = 900,000$, 60,747 times Proposition 2’s
148 bound at $T = 10^6$ (Table 1, App. A); (2) makes that an if-and-only-if against the whole legal
149 adversary class, not a pattern across five settings. **The claim is that necessity: a tightness result**
150 **about someone else’s sufficient condition**, smaller and more precise than a forfeit of the rate by
151 post-processing. Elsewhere it stays sufficient only, and now provably so: the tangent integrator above
152 forfeits no coverage at any width Corollary 2 permits. At $\hat{q} \equiv +b/2$ the looseness goes the other way
153 and disappears: Corollary 2 permits radius 0 and (2) no width at all against the mirror adversary, the
154 unsmoothed control included, though $\tau = 1.5$ covers there against the specified one. That asymmetry
155 makes “tight” a located claim: a tie-convention property of the cited closed statements.

156 **The boundary read as a bet.** The boundary has a betting reading (Appendix B): under a
157 Bernoulli(α) null on err_t , imposed for that reading alone and on nothing else here, an anytime-
158 valid test convicts the $\tau = 1.5$ arm past the edge at the 5% level and cannot reject inside it at any
159 horizon. It adds a diagnostic and a reading, not a result, and no result above assumes that null or is
160 certified by that test.

161 4 Limitations

162 **The retention direction is proved with no gap only where the two boundaries meet:** a constant
163 scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run
164 here; a genuinely time-varying $\hat{q}$ leaves the band $\sigma^+ < \tau \leq \tau^{*+}$ open in neither direction (Section 3).
165 And the excursion is specific to the saturator’s *level*, which condition (4) bounds only from below.
166 **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$,
167 the certified gap $(ch(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was
168 not a tight certificate. **The dead-band widths are absolute, and the failure is total only against**
169 **the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75b$. Under i.i.d. scores it
170 returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat.
171 The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly.
172 **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done
173 without sight of them. **Placement B is conceded, not tested.** **The i.i.d. check above used one seed**
174 **(20260820, $T = 10^6$); the primary, adversarial harness uses none, and only its null-scorecaster**
175 **edge is bracketed;** the others are one-sided. **What remains open is the band between the two**
176 **boundaries.** For a genuinely time-varying $\hat{q}$, coverage in $\sigma^+ < \tau \leq \tau^{*+}$ is neither proved nor
177 falsified; Table 2’s power-of-two rows sit there, and closing that band is future work, not a sweep this
178 paper has already run.

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282 Code and data

283 The simulator, its frozen configuration, the results/ files behind every figure and table, the gen-
284 erators that read them, and this project’s audit trail across sessions are at [https://\[REPOSITORY](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION])
285 [LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION\]](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION]).

286 A Supporting measurements

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. Proposition 2’s bound is 13.2061 at $T = 2 \times 10^5$ and 14.8155 at $T = 10^6$; “ratio” is $\max_t |E_t|$ over it, and “Sat.” the fraction of rounds on which (4) delivers its full $\pm b$. Every number is measured against the adversary Section 2 specifies.

Readout on the completed threshold	miscov. 2×10^5	miscov. 10^6	$\max_t E_t 10^6$	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	0.100004	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	1.000000	1.000000	900,000	60,747	1.0000	0.5

Table 2: Legal $(r_t, \hat{q})$ configurations on which the printed τ^* over-certifies, each under two legal adversaries; “proved window” is where (2) certifies retention against every legal adversary, and between it and the failure boundaries $\tau^{*\pm}$ neither direction is claimed, which is where the rows built on a supremum the dynamics cannot reach or never attain sit. Here $\emptyset$ marks a configuration for which no width is certified, $\tau = 0$ included, and $T = 10^5$ except the two power-of-two rows, at $T = 10^6$. The last row carries no dead band at all.

$(r_t, \hat{q})$ configuration	τ	printed τ^*	proved window	$s_t = \hat{q}_t + \varepsilon$		$s_t \equiv -b/2$	
				miscov.	$\max_t E_t $	miscov.	$\max_t E_t $
$\hat{q} \equiv +0.4$, level b	0.599	1.4	$\tau < 0.6$	0.100010	7.20	0.099890	11.60
$\hat{q} \equiv +0.4$, level b	0.8	1.4	$\tau < 0.6$	0.100050	8.70	0.000000	10,000
$\sup_x r_t = 10b$, off the reach	1.001	19	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t$ unbounded, off the reach	1.5	∞	$\tau < 1$	1.000000	90,000	0.000000	10,000
$A^+ = 4b$, $A^- = -b$	1.5	7	$\tau < 1$	0.100020	4.30	0.000000	10,000
level $4b$, $\hat{q} \equiv 0$	6.999	7	$\tau < 7$	0.100070	12.20	0.100050	11.60
level $4b$, $\hat{q} \equiv 0$	7.0	7	$\tau < 7$	0.100090	12.20	0.000000	10,000
$\sup_x r_t = 2b$, unattained	2.99	3	$\tau < 1$	0.100320	211.10	0.098010	210.50
$\sup_x r_t = 2b$, unattained	3.0	3	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t = 2b$, attained	3.0	3	$\tau < 3$	0.100000	12.20	0.000000	10,000
$\hat{q}_t = b/2$ on powers of 2	1.0	2	$\emptyset$	0.100009	14.30	0.000000	100,000
$\hat{q}_t = b/2$ on powers of 2	1.9	2	$\emptyset$	0.110960	84,745.70	0.000000	100,000
$\hat{q} \equiv +b/2$, no band	0	2	$\emptyset$	0.100000	0.90	0.000000	10,000

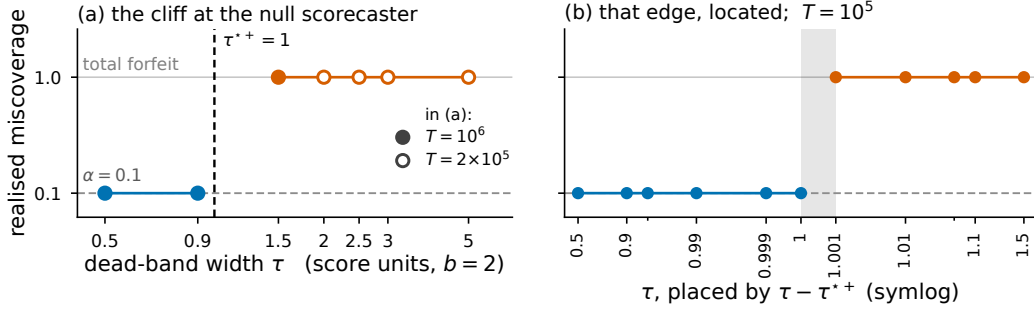


Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b , under the *specified* adversary, where the upward boundary τ^{*+} of (2) is 1: **(a)** the full sweep, filled markers to $T = 10^6$ and the four widest bands, drawn open, to $T = 2 \times 10^5$, blue covering and orange forfeiting, with the dashed rule at τ^{*+} and the grey band spanning the widest covering width to the narrowest failing one. **(b)** The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^{*+}$ on a symlog axis (linear zone $|\tau - \tau^{*+}| \leq 10^{-3}$; unlabelled minor ticks at $\tau = 0.95$ and 1.05) that magnifies the neighbourhood of τ^{*+} , so visual spacing there is not proportional to true distance in τ . Figure 3 gives the other four $(r_t, \hat{q})$ settings.

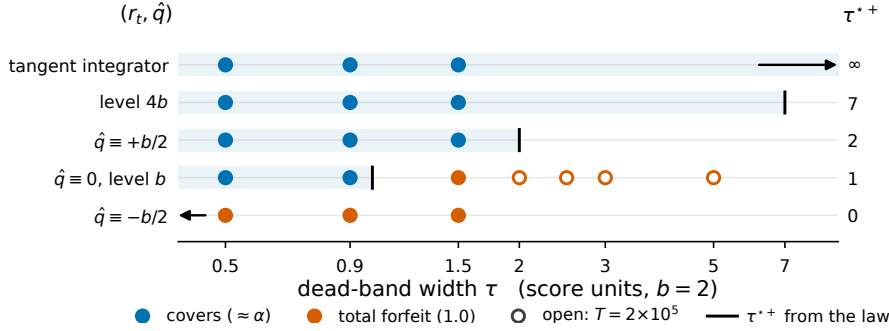


Figure 3: All 19 dead-band runs at the five admissible $(r_t, \hat{q})$ settings, rows ordered by τ^* on the shared horizontal τ axis: blue markers cover at $\approx \alpha$ and orange markers return 1.000000, filled to $T = 10^6$ and open to $T = 2 \times 10^5$, all under the *specified* adversary. The rule on each row is that setting's upward boundary $\tau^{*+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, with the tinted band $\tau \leq \tau^{*+}$; two rules fall off the axis and are drawn as arrows, $\tau^{*+} = 0$ at $\hat{q} \equiv -b/2$ and τ^{*+} unbounded under Angelopoulos et al.'s tangent integrator. The mirror boundary τ^{*-} of (2) is not drawn here and differs from τ^{*+} at the two non-null scorecasters.

287 B The betting reading

288 The boundary is stated in $|E_T|$, which is the net gain of a Skeptic who pays α each round for err_t ;
 289 that is what gives it a betting reading. The reading Section 3 points at is Ville's inequality applied
 290 to the standard martingale $M_t(\lambda) = \prod_{i \leq t} (1 + \lambda(\text{err}_i - \alpha))$, whose logarithm depends on the path
 291 only through t and E_t [Shafer and Vovk, 2019, Ramdas et al., 2023, Vovk and Wang, 2021], under
 292 the imposed null that $\text{err}_t | \mathcal{F}_{t-1}$ is Bernoulli(α), with $\mathcal{F}_{t-1}$ the history through round $t-1$. Ville's
 293 inequality needs each bet to be predictable, fixed by the history before the round it is scored on,
 294 so the direction of failure cannot be read off the realised path first. The diagnostic is therefore the
 295 two-sided mixture $M_t = \frac{1}{2} M_t(+1) + \frac{1}{2} M_t(-1)$, which is a test martingale for the same null because
 296 a convex combination of test martingales is one, and which fixes no direction in advance; both signs
 297 lie in $[-1/(1-\alpha), 1/\alpha]$, the range on which $M_t(\lambda) \geq 0$ on every path, the magnitude 1 is the same
 298 for every arm and both directions, and the mixture costs a factor of two against a correctly guessed
 299 direction. So read, an anytime-valid test of the deployed intervals' calibration convicts the $\tau = 1.5$
 300 arm on its sixth interval at the 5% level, where $M_5 = 12.38$ and $M_6 = 23.52$ against the rejection
 301 threshold $1/0.05 = 20$. Inside the edge the same test cannot reject at that level at any horizon: it
 302 peaks at 2.03 for $\tau = 0.5$ and 3.92 for $\tau = 0.9$, and from round 62 on Proposition 2's bound on
 303 $|E_t|$ forbids rejection on every retained trajectory. M_t is an e-value for that imposed null alone; the

304 accumulator itself is signed and is neither an e-value nor an e-process, and no result in Section 3
 305 assumes that null or is certified by that test.

306 C Where this sits

307 **One quantity, four names.** Four literatures name the movement of a sequentially updated fore-
 308 cast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC,
 309 MQC/SMQC, congruence, nervousness); *jumpiness* [Zsótér et al., 2009], *convergence index* [Ehret,
 310 2010], *(in)consistency* [Pappenberger et al., 2011]; and *switching cost*. The same arithmetic is the
 311 Multi-Quantile Change over a predictive distribution [Zanotti, 2026, v3, §3.4.1], IACER in applied
 312 control [Sarhadi, 2025], and a path length in dynamic-regret online learning. Online conformal
 313 prediction gives distribution-free long-run coverage for any bounded scorecaster [Angelopoulos et al.,
 314 2023, Thm 1, Prop. 2], generalizing the running-error feedback that adaptive conformal inference
 315 [Gibbs and Candès, 2021] and its distribution-shift variant DtACI [Gibbs and Candès, 2024] already
 316 apply directly to the threshold. Forecast stability treats the movement as a functional, read out as
 317 post-processing [Godahewa et al., 2025, Eq. 3], against a priced decision to switch [Genov et al.,
 318 2026, §4.2], [Van Belle et al., 2026]. Verification work uses a matched-level design against real
 319 producers from 2012 [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning
 320 gives worst-case theory for the penalised problem [Andrew et al., 2013, Thm 2].

321 The object that settles a deployed interval’s validity (the admissible set around the deployed threshold)
 322 sits in the first of them. Those four vocabularies are where this paper’s claim and its concessions have
 323 to be placed. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023,
 324 Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster, and run
 325 a smoothing-family model in that slot themselves; this paper proves that admissible set tight. Dupuy
 326 et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster.
 327 The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2]
 328 prove the retention statement in Conformal PID’s own notation, with Wang and Hyndman [2024] an
 329 independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both
 330 readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a
 331 smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a
 332 feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al.,
 333 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model*
 334 update on the conformal score. Their one non-standard assumption is that $\sum_{t \leq T} q_t \leq O(\alpha h(T))$
 335 for a sublinear h , supported with a plot rather than derived. Verification’s matched-level design tests
 336 real producers from 2012, with a control stronger than the pair: the whole marginal [Pinson and
 337 Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning adds a Thm. 7 trade-off in which
 338 sublinear regret is bought by growing θ with T , which grows the ratio [Andrew et al., 2013, Thm. 7].

339 NeurIPS Paper Checklist

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341 Question: Do the main claims made in the abstract and introduction accurately reflect the
 342 paper’s contributions and scope?

343 Answer: [\[Yes\]](#)

344 Justification: The abstract and Section 1 state exactly two claims and no more: that this
 345 paper joins an already-refereed claim to its already-published refutation (Conformal PID’s
 346 scorecaster slot vs. the deployed-threshold Corollary 2), and that the admissible radius
 347 Corollary 2 relies on is *tight* at the null scorecaster paired with the minimal admissible
 348 saturator, proved as a necessary-and-sufficient condition on that sub-class in Section 3. Both
 349 the abstract’s hedges (“elsewhere the condition is sufficient only”, the two degenerate-case
 350 caveats) and Section 4’s concessions match the proved scope; no claim in the abstract
 351 exceeds what Section 3’s theorem and Section 4 state.

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 354 made in the paper.

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Question: Does the paper discuss the limitations of the work performed by the authors?

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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no gap on the sub-class every tested setting inhabits; the closed form (3) is stated as an if-and-only-if with its two enabling hypotheses named explicitly as load-bearing rather than decorative. **One honest caveat:** the proof is presented in narrative prose within the section text, with assumptions stated in the surrounding sentences and conditions given as labelled equations, rather than as a numbered theorem/proof LaTeX environment with a formal statement–proof separation. The mathematical content (assumptions, both proof directions, both degenerate edge cases) is complete and was independently re-derived and adversarially attacked in this project’s own process; converting the presentation to a numbered theorem environment is a formatting change deferred to a future revision, not a gap in the argument.

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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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- The full details can be provided either with the code, in appendix, or as supplemental material.

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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